

LOCK-FREE SEQLOCK BOUNDARY CONTRACT & PREEMPTION RECOVERY

Asynchronous Inter-Core State Exchange with Zero Wait States and Deterministic Fallback Hierarchy

(a) Lock-Free Seqlock Publication Protocol

Atomic Barriers · Zero Mutex

UNTRUSTED WRITER (NPU/MPU)

1. `atomic_store_release(S, 2k+1)`

Marks sequence counter ODD

2. Store-Release Barrier

Flushes counter visibility

3. Write Payload Fields

`m_cmd`, `k`, `t_prod`, `Δt_valid`, CRC

4. Store-Release Barrier

Ensures payload commits first

5. `atomic_store_release(S, 2k+2)`

Marks sequence counter EVEN

SAFETY READER (Real-Time MCU)

1. `S1 = atomic_load_acquire(S)`

Sample initial sequence

2. `if (S1 & 1) reject();`

Odd → Write in progress! Zero wait

3. Load-Acquire & Copy

Copy payload to private stack

4. `S2 = atomic_load_acquire(S)`

Sample second sequence

5. `if (S1 != S2) reject_torn();`

Torn write trap (zero blocking)

SELF-CONTAINED BOUNDARY CONTRACT PAYLOAD (256 bytes)

<code>dot_m_cmd</code>	[float32]	Commanded mass flow target (physical limits [0, 45 kg/s])
<code>seq_id (k)</code>	[uint64]	Monotonic proposal index (detects dropped/skipped frames)
<code>t_prod</code>	[uint64]	Monotonic hardware timestamp of inference completion
<code>Δt_valid</code>	[uint32]	Lease expiration duration (e.g. 45.0 ms maximum validity)
<code>crc32_hash</code>	[uint32]	Payload integrity checksum (detects corrupted bitflips)

(b) Preemption Points & Deterministic Fallbacks

Guaranteed Bounded Response

WRITER PREEMPTION / CRASH POINT ANALYSIS

Point A · Crash Before Step 1

[S is even (2k)]

Reader sees intact prior record (2k==2k). Zero corruption. ✓

Point B · Preempted During Write

[S is odd (2k+1)]

Reader tests (S1 & 1) != 0, drops buffer in <5 μs. Zero blocking. ✓

Point C · Crashed Mid-Publication

[S stays odd]

Reader detects persistent odd sequence, trips fallback clamp. ✓

Point D · Write Fully Committed

[S is even (2k+2)]

Reader validates $S1 == S2 == 2k+2$ and CRC. Accepts fresh packet. ✓

4 BOUNDARY FAILURE CLASSES & DETERMINISTIC RESPONSES

1. MALFORMED: CRC mismatch / Out-of-bounds `dot_m_cmd`

Fallback Action: Clamp valve immediately to last verified safe orifice position.

2. STALE: `t_now > t_prod + Δt_valid` (Lease expired)

Fallback Action: Hold position for $t_{hold} = 10$ ms; if persists, begin ramp-down.

3. MISSING: Sequence gap (`k_new > k_prev + 1`)

Fallback Action: Flag queue-overflow; apply state extrapolation with rate limit.

4. TORN / OVERRUN: Seqlock mismatch (`S1 != S2`)

Fallback Action: Reject torn read instantly (<5 μs); continue 1 kHz control tick.